

Medical Record Science I

BBAHM 202 (Major) | Complete All-Modules Summary | 2024-2025 Sem 2

5 Credits (4T + 1P) | Maulana Abul Kalam Azad University of Technology, West Bengal

MODULE M1 — Medical Records Basics

Historical Background

- First medical record unit: **1667, St. Bartholomew's Hospital, England.**
- Pennsylvania Hospital, USA began a patient register in **1792.**
- In India: **Bhore Committee (1946)** first stressed importance; reiterated by **Mudaliar Committee (1962).**
- Jain Committee and Rao Committee highlighted poor records and recommended proper Medical Records Section in each hospital.

Definition of Medical Records

- An **orderly, written document** encompassing patient's health history, physical examination, findings, lab reports, treatment, surgical procedure reports, and hospital course.
- A **legal document** providing a chronicle of a patient's medical history and care.
- Must justify: investigations, diagnosis, treatment, length of stay, results of care, and future course of action.

Role of Medical Records in Healthcare Delivery

For the Patient: Documents clinical history; avoids repetition of diagnostic measures; assists continuity of care; provides data for insurance, employment, and health schemes.

For the Doctor: Assures quality and adequacy of diagnosis/treatment; aids continuity of care; aids research and education; protects in legal matters.

For the Hospital: Documents type and quantity of work; evaluates physician proficiency; evaluates services against norms; protects hospital legally; supports administrative decisions (staffing, budget, planning).

For Medical Education and Research: Basis for all clinical research; furthers education of doctors and health personnel; supplies data to public health authorities for disease control.

Types of Medical Records

Type	Description
Outpatient (OPD)	OPD card (9"×6"), identification data, history, physical findings, diagnosis, treatment, follow-up.
Inpatient (IPD)	Initiated at admitting office; includes all clinical documents from admission to discharge.
Emergency (A&E;)	Registration, casualty number, MLC registration if applicable.
Diagnostic	Laboratory, radiology, pathology, and other investigation reports.

Type	Description
Surgical / Operation	Anaesthesia record, operation record, post-operative notes.
Medico-Legal (MLC)	Cases with legal implications — accidents, assault, poisoning, sexual offences, etc.
Administrative	Census reports, statistical reports, personnel records.

Format Types of Medical Records

- **Source-Oriented Medical Record (SOMR):** Documents organized by source — physician notes, nursing notes, lab, radiology. Traditional method.
- **Problem-Oriented Medical Record (POMR):** Documents organized around patient problems. Introduced by Dr. Lawrence Weed. Uses SOAP notes.

Characteristics of a Good Medical Record

Characteristic	Meaning
Complete	Sufficient data to identify patient; justify diagnosis, treatment, follow-up, and outcome.
Adequate	All necessary forms; all clinical information present.
Accurate	Factually correct; capable of quantitative analysis.
Legible	Readable by all authorized personnel.
Timely	Entered promptly at the time of care.
Confidential	Protected from unauthorized access.
Retrievable	Easy to locate and access when needed.
Standardized	Follows uniform formats and conventions.

Ownership of Medical Records

- Medical records are the **property of the hospital/healthcare facility** that created them.
- The **patient has the right to access** information within their own records.
- The hospital is the **custodian**; the patient holds the right to information.
- Records cannot be released to third parties without the patient's written consent, except by court order or public health mandate.

Contents of Medical Records

- Front sheet / Identification summary sheet (face sheet)
- Consent for treatment
- Legal documents (referral letter, request for information, etc.)
- Admission notes
- History and physical examination findings
- Clinical progress notes (doctor's and nurse's)
- Orders for treatment and medication forms
- Laboratory and investigation reports (X-ray, pathology, etc.)
- Anaesthesia record (if surgery performed)

- Operation report
- Temperature chart
- Discharge summary / referral slip

Maintenance of Records in the Ward

- Done by **ward nurse or ward clerk (AMRT — Assistant Medical Records Technician)**.
- Functions: registration of admissions/discharges; mounting investigation reports; bed position accounts; scheduling follow-up appointments; daily ward census reports.
- Admission check desk: receives admission advice; creates/updates patient index card; makes entries in accession register.
- Census desk: prepares admission and discharge lists; collects discharged patient records daily; prepares census reports.

Flow Chart of Functions — Inpatient Medical Record

Patient Flow: Registration → Consultant → IP Admission → Ward → On Discharge

MRD Processing: Assembling → Deficiency Check → Coding → Indexing → Computer Entry → Scanning → Permanent Filing → Retrieval

Assembling and Deficiency Check

Standard Assembly Order:

1. Face sheet
2. Case summary and discharge records
3. History and physical examination
4. Labour record (if applicable)
5. Consultation record
6. Clinical progress notes
7. Laboratory and investigation reports
8. Anaesthesia record
9. Operation record
10. Nurses' records
11. Temperature chart
12. OPD card

Deficiency Check Process:

- Check for: missing signatures, final diagnoses, investigation reports, consent forms.
- Complete records → permanent filing area.
- Incomplete records → deficiency slip → sent to doctors for completion.
- List of incomplete records submitted periodically to hospital administration.

MODULE M2 — Coding and Record Management

Medical Coding — Definition and Purpose

- Process of **translating diagnoses, procedures, and supply information** from patient records into universal alphanumeric codes.

- Used for: billing, statistics, research, insurance reimbursement, public health reporting.
- Medical coders must be knowledgeable in medical terminology and anatomy.

Types of Medical Codes

Code System	Purpose
ICD (ICD-9 / ICD-10 / ICD-11)	Describe illness, injury, or cause of death.
CPT — Current Procedural Terminology	Surgeries, radiology, anaesthesia, measurement procedures.
HCPCS Level 2	Outpatient hospital care and medical aid.

6 Key Steps in the Medical Coding Process

1. **Abstract the documentation:** Read all physician notes completely — procedure notes, operative reports, history & physical. Understand Who, Where, Why, What, How.
2. **Query if necessary:** Ask for clarification if documentation is unclear. "If it's not documented, it didn't happen." Legal queries must ask open-ended questions.
3. **Code the diagnosis/diagnoses:** Select most accurate, most specific code. Review inclusive signs and symptoms.
4. **Code the procedure/procedures:** Assign codes for all procedures. Note inclusive components (e.g., simple repair after excision is included in the Excision code).
5. **Confirm medical necessity:** Align diagnosis codes with procedure codes to justify treatment. Essential for reimbursement and legal/ethical review.
6. **Double-check codes:** Code backwards — look up code, re-read description, compare with documentation.

ICD — International Classification of Diseases

- WHO's standardized diagnostic classification system assigning unique codes to diseases, injuries, and health conditions.
- Objectives: standardization of health information; disease surveillance; mortality/morbidity statistics; health planning; research; insurance billing.

Feature	ICD-9	ICD-10	ICD-11
Year	1975	1990	2019 (effective 2022)
Code Format	Numeric (3–5 digits)	Alphanumeric (1 letter + digits)	Alphanumeric (fully digital)
No. of Codes	~13,000	~68,000+	~55,000+ (expandable)
EHR Compatible	No	Partially	Yes — fully
Specificity	Low	High	Very High
Example	250.0 = Diabetes	E11 = Type 2 DM	5A11 = Type 2 DM

POMR — Problem-Oriented Medical Record

- Introduced by **Dr. Lawrence Weed**.
- A structured method focused on patient's **problems** and their clinical evolution; allows systematic, organized view of patient history.

4 Phases of POMR:

1. Formation of a **Database** (history, physical exam, lab results).
2. Identification of a **Problem List** — active and inactive problems.
3. Identification of a specific **Plan of Care (POC)** for each problem; includes SOAP progress notes.
4. Determination of **effectiveness of POC** and changes based on patient progress.

5 Basic Components of POMR:

Component	Description
1. Database	Complete history, physical exam, and lab data. Additional information added as collected.
2. Problem List	Constructed after database. Problems are active (current) or inactive (prior, resolved but still important).
3. Initial Plans	Written by admitting physician after problem list. States "what to do about what is wrong."
4. Daily Progress Notes	Written in SOAP format for each active problem. Not required every day if nothing has changed.
5. Discharge Summary	All active problems resolved to their furthest point. S, O, A, P for each problem. Emphasis on unresolved problems.

SOAP Format — Progress Notes:

Letter	Meaning	Description
S	Subjective	What the patient reports; abnormalities from history pertaining to each problem.
O	Objective	Measurable physical examination and lab findings.
A	Assessment	Physician's interpretation of the problem; differential diagnosis in prioritized manner if needed.
P	Plan	1. Diagnostic Plan — rule in/out each differential. 2. Therapeutic Plan — initial therapies and rationale. 3. Patient Education Plan — educate patient about problem and future management.

Indexing

- Process of maintaining or arranging patient medical reports in a systematic order to **locate specific documents quickly**.

Index Type	Description
Alphabetic / Master Patient Index (MPI)	Indexed by patient name alphabetically. Provides entry into filing system. Used when MR number is not known.
Disease Index	Indexed by ICD diagnosis code. Cards (3"×5") with patient data, age, sex, treatment result, complications.
Operation / Procedure Index	Indexed by surgical/medical procedures performed.
Physician Index	Indexed by treating doctor; used to evaluate physician performance.
Unit Index	Details of patients treated in a specific unit; used to evaluate unit performance.

Medical Record Numbering Systems

System	Description	Key Feature
Serial	New number every visit.	Multiple numbers for same patient.
Unit	One number on first visit; same for all subsequent visits.	One composite record per patient.
Serial Unit	New number each visit; old records brought forward under latest number.	Tracer card left at old number pointing to new.
Annual	Year digits added to end of serial number.	Controls inactivation of records by year.
Family	Extra two digits for family member: 01=Head, 02=Spouse, 03=Children, 04=Others.	Used in family health programmes.

Filing Systems

Method	Description	Advantage	Disadvantage
Straight Numeric	Records in exact ascending order (65023, 65024...).	Easiest; simplest training.	Easy to misfile; filing congestion at one end.
Terminal Digit	6-digit no. split into 3 pairs. Primary = LAST two digits; Secondary = middle two; Tertiary = first two.	Even distribution across 100 sections; misfiling greatly reduced; clerks assigned specific sections.	Slightly longer training period.
Middle Digit	Primary = MIDDLE two digits; Secondary = left two; Tertiary = right two.	Easy to pull 100 consecutive records; simpler conversion from straight numeric.	More training required.
Alphabetical	Filed by patient name alphabetically.	Simple for small hospitals.	Difficult in large hospitals; error-prone with common names.

Colour Coding of Folders

- Colour bars on folder edges create distinct visual patterns in each file section.
- A break in the colour pattern immediately signals a misfiled record.
- Most effective with terminal digit or middle digit filing.
- Ten different colours for primary digits 0–9. Two colour bars signify each of the two primary digits.

Microfilming

- Saving copies of records in microscopic form. Used in large hospitals due to space constraints.
- Saves storage space by up to **90%**. If stored properly, microfilm can last **hundreds of years**.

Type	Description
Roll Film	On a reel; 16 mm and 35 mm formats.
Microfiche — COM	Grid format; holds up to 300 pages of images.
Microfiche — Jacket	Camera film in plastic jackets; 5 images across.

Type	Description
Aperture Cards	Timecard-like with hole for film; holds one image; used in engineering.
Advantages	Disadvantages
• 90% space saving • Lasts hundreds of years (if stored properly) • Low cost vs. computers/hard drives • Eliminates misfiling • Easy accessibility and retrieval	• Slow retrieval rate • Hard to update • Only one person can view a roll at a time • Deteriorates if exposed to heat, humidity, light, or chemicals

Ideal Storage Conditions: Sealed containers in non-corrosive metal cabinets; avoid heat, humidity, light, and chemicals. Create duplicate backup copies or digitize.

Computerization — EHR vs. EMR

EMR (Electronic Medical Record)	EHR (Electronic Health Record)
Digital record from a single provider/clinic.	Accessible anywhere; shareable across providers.
Stored locally; not intended for distribution outside individual practice.	Cloud-based/networked; enables real-time collaborative care.
Designed for one practice's internal use.	Provides comprehensive healthcare history accessible by authorized users.

Hospital Statistics — Key Formulas

Bed Occupancy Rate (BOR) = (Occupied Bed Days ÷ Available Bed Days) × 100

Average Length of Stay (ALOS) = Total Inpatient Days ÷ Total Discharges (incl. deaths)

Gross Death Rate = (Total Deaths ÷ Total Discharges incl. deaths) × 100

Net Death Rate = (Deaths occurring after 48 hrs of admission ÷ Total Discharges) × 100

Bed Turnover Rate = Total Discharges ÷ Average Beds Available

OPD / IPD Ratio = Total OPD Patients ÷ Total IPD Admissions

Uses of Hospital Statistics: Hospital planning and resource allocation; evaluation of quality of care; government reporting and compliance; research and epidemiological studies; identifying deficiencies in staff, facilities, and methods.

Process of Arranging Medical Records

- Collection of daily ward census reports along with discharged patient records.
- Analysis and assembly in prescribed standard order; deficiency check.
- Coding and indexing as per latest ICD revision.
- Computer entry → scanning → permanent filing → retrieval as needed.
- **Reasons for retrieval:** patient follow-up; ward/casualty re-admission; research; medical reimbursement claims; production in court of law.

MODULE M3 — Medical Record Department (MRD)

Goal of MRD

- To maintain a **modern and scientific** medical record system in conformity with international standards.
- To ensure records contain **comprehensive and accurate social, medical, and vital information** to facilitate effective medical services.

Organizational Structure

MRO → Asst. MRO (×2) → MRTs → Sections: OPD (CR & Appointment / OP Clinic) | A&E | Admissions | Ward | Medical Secretary | Processing OP/IP Records | Statistics | Filing & Retrieval (Records & X-rays) | General

Four Functional Categories of MRD

Section	Key Functions
OPD — Central Registration (3 shifts, 24×7)	Control of hospital numbers; registration of new cases; MPI maintenance; follow-up appointment registration; supply records to OPD clinics.
OPD — Outpatient Clinic	Collection of patient records and X-rays; maintenance of patient treatment accounts; collection and return of records to files.
A&E; / Casualty (3 shifts, 24×7)	Control of casualty numbers; registration of A&E; and MLC cases; mounting investigation reports; collection of statistics and filing A&E; records.
Admissions Office (3 shifts, 24×7)	Maintenance of waiting list; registration of admissions/discharges; bed occupancy board; securing patient property.
Ward	Registration; mounting investigation reports; bed position accounts; scheduling follow-up appointments; daily census reports.
Medical Record Library (3 shifts, 24×7)	Processing OP/IP records; hospital statistics collection and analysis; filing and retrieval of records and X-rays.

Staffing Guidelines

Formula: $(\text{Working Days} \times \text{Number of Shifts}) \div \text{Man-Days} = \text{Persons Required}$ Example: $(365 \times 3) \div 217 = 5$ persons per unit (Man-days = working days after subtracting holidays)

Hospital Size	Personnel Formula	Total Personnel
100 beds	2 (base minimum) + 1/100 beds + 1/10 beds	2 + 1 + 10 = 13
500 beds	2 + 5 + 50	57

Duties and Responsibilities of Medical Record Officer (MRO)

- Establish, organize, and manage the MRD effectively.
- Develop policies and procedures as per the Health Directorate/Ministry of Health.
- Assist the Medical Record Committee in developing different forms.
- Review medical records of different departments.
- Participate in quality assurance, utilization review, infection control, and other committees.
- Prepare medical reports, certificates, birth/death registers; notify concerned authorities.
- Prepare monthly statistical reports on hospital activities.
- Observe professional ethics and protect confidentiality of patient information.
- Select appropriate MRD personnel; prepare educational training programmes.
- Design and develop standard basic medical record forms.

- Maintain effective Master Patient Index.
- Carry out quantitative and qualitative analysis of records (concurrent and retrospective).
- Establish "Incomplete Medical Records Control" — review all discharged records promptly.
- Create "Doctors Conference Room" in MRD for completing deficient records.
- Submit regular reports on delinquent records with respective doctors' names.

Role of Medical Record Technician (MRT)

- Coding diagnoses using ICD; assembling discharged patient records in standard order.
- Deficiency checking; indexing records (disease, procedure, physician, patient).
- Filing and retrieval of records and X-rays; collection and compilation of hospital statistics.
- Medico-legal documentation assistance; registration of new patients; MPI maintenance.

Retention of Medical Records

Category	Retention Period
Adult patients — general	5–10 years from date of last visit
Minors (children)	Until age of majority + specified years
Medico-Legal Cases (MLC)	10+ years (as per legal requirements)
Disease index / Operative index cards	10 years
Statistical data	5 years
Death records	Permanent
Birth records	Permanent

Factors Determining Retention Period:

- Internal organizational need.
- Regulatory requirements for inspection or audit.
- Legal statutes of limitation and involvement in litigation.
- Taxation and financial reporting needs; mandatory state statutes.

Preservation of Medical Records

Manual Preservation Methods:

Method	Key Details
1. Paper and Ink Selection	Use specific ink and paper for records lasting 10+ years. USA and UK have developed special materials for long-term preservation.
2. Protection from Decay and Rot	Ensure proper cross-ventilation; adequate fans or air-conditioning; regular vacuuming. Avoid hot, dry climate; stagnant air; direct sunlight; dust; dampness.
3. Protection from Insects	Insecticidal powders/sprays; naphthalene balls in bookshelves; fumigation (non-damaging fumigants). Use closed steel racks (not wood — attracts white ants). Racks 5" from damp walls.

Method	Key Details
4. Atmospheric Pollution	Acidic gases (SO ₂) in cities damage records faster. Use air-conditioning with alkaline water treatment (pH ≥ 8.6) to neutralize acidic gases.
5. Fire Safety	No smoking or open flames in record room. Adequate fire extinguishers. All electric wires through conduits. Wire-net frames on windows and ventilators.
6. Temperature and Humidity	Ideal: 22–25°C and Relative Humidity 45–55%. Temperature >32°C or humidity >70% causes growth of pests (silverfish, cockroaches, termites, fungi). Use commercial humidifiers.
7. Microfilming Control Register	Maintained with: hospital number, patient name, age, sex, date last admitted, date microfilmed, microfilm file number, date original destroyed, and remarks.

Electronic Preservation Methods:

- HIS (Hospital Information System), EMR, EHR, PACS (Picture Archiving and Communication System — for radiology).
- Scanning and digitization of paper records; cloud storage; RAID (Redundant Array of Independent Disks).
- Optical discs; encryption for data security; automated backup systems.

Destruction Procedure

- Inactive records destroyed as per retention schedule.
- Methods of destruction: shredding or burning.
- A **Destruction Register** must be maintained — records: hospital number, patient name, age, sex, address, registration date, admission status, final diagnosis, and operations before destruction.
- Written permission must be obtained from hospital administration; all departments must be informed before destruction.

MODULE M4 — Medical Record Administration & Legal Aspects

Medico-Legal Cases (MLC) — Definition

- A case of injury or ailment in which **investigation by law enforcement agencies is essential** to fix responsibility regarding the causation of the said injury or ailment.
- Involves both **medical jurisprudence** (branch of medicine) and **medical law** (branch of law).

Incidents that Come Under MLC

Category	Examples
Trauma and Violence	Assault, battery, domestic violence, child abuse, Road Traffic Accidents (RTA), industrial accidents, stab injuries, firearm injuries.
Poisoning and Chemical	Poisoning, drug overdose, alcohol intoxication, chemical injuries, electrical injuries.
Sexual Offences	Sexual offences, criminal abortions.
Asphyxia	Hanging, drowning, suffocation, strangulation.
Burns	Burns and scalds.

Category	Examples
Deaths	Custodial deaths; death in OT; unnatural deaths; death by snake/animal bite; brought dead (Found Dead); death within 24 hrs of hospitalization without establishment of diagnosis.
Others	Undiagnosed coma; attempted suicide.

Police Intimation (Under Section 39, CrPC)

- Attending Medical Officer is **legally bound** to inform police about arrival of an MLC.
- Failure to report may invite prosecution under **Sections 176 and/or 202 of IPC**.
- 3 copies of police intimation: 1st → nearest police station; 2nd → police station of incident location; 3rd → medical records.
- Police must be informed on discharge, transfer, or death of the patient.

Documentation of MLC

- Done in **duplicate** in a set Proforma; legible handwriting; complete words; no abbreviations or shortcuts.
- No cutting or overwriting; all corrections properly initialled.
- All columns filled by the **same doctor** who examined the patient; completed then and there — no provisions left to fill later.
- Investigation findings, treatment given, and opinion recorded in MLC sheet. If opinion not possible: write "under observation" and sign.
- After all investigations: same doctor gives final opinion in original MLC sheet; signs with full name and designation.
- All documents after MLC registration bear the same MLC number, including the discharge slip.
- All MLC X-rays kept in department as evidence for court of law.
- If patient dies: handed over to police for post-mortem (not to relatives). Copy of MLC sheet given to police against their signature.

MLC Form — General Details Recorded

Registration No. | MLC No. | Name | S/D/W of | Age | Sex | Religion | Occupation | Residential Address | Brought By | Date & Time of Examination | Name of Police Station

Evidence Preservation by Case Type

Case Type	Evidence to Preserve
Bullet / Stab / Burn Injury	Clothing with tears, holes, cuts, blood stains. Bullets recovered from body must be marked by etching initials before preservation. Each piece of evidence circled and numbered with matching description in MLC report.
Poisoning Cases	Gastric lavage / gastric contents / vomitus; soiled clothing; blood, urine, and any other relevant body fluid depending on poison ingested.
Sexual Offences	Clothing with blood stains, seminal stains, or tears/cuts. Vaginal swab.

Transfer of MLCs

- Transferring hospital must provide treatment within its capacity to minimize risk to patient.
- Receiving hospital must have space, qualified personnel, and must have agreed to accept the transfer.

- All medical records (history, examination findings, diagnostic results, provisional diagnosis, treatment) transferred with patient.
- Written informed consent/certification required. Transfer via qualified personnel with life support equipment.

Brought Dead Cases

- All cases of unnatural death (accidents, burns, violence, sexual assault) must be labelled as medico-legal; police constable on duty must be informed.
- **In death certificate:** cause of death = "NOT KNOWN — BROUGHT DEAD."
- NEVER write that cause of death should be determined after post-mortem in the death certificate.

Legal Documents in Medical Records

Document	Purpose
Consent Forms	Informed consent for treatment, surgery, or procedures.
MLC Register	Official record of all medico-legal cases registered.
Post-mortem Report	Forensic report on cause of death.
Wound Certificate	Documentation of injuries for legal purposes.
Birth Certificate	Official registration of births.
Death Certificate	Official registration of deaths with cause.
DAMA Form	Discharge Against Medical Advice — records patient's refusal of further treatment.
Referral / Court Summons	Referral letters, requests for information, responses to court orders.

Patient Rights

- Right to access own medical records.
- Right to confidentiality of personal health information.
- Right to informed consent before any procedure.
- Right to correction of errors in records.
- Right to refuse treatment (Discharge Against Medical Advice — DAMA).

Confidentiality

- Patient information cannot be disclosed without written consent **EXCEPT:**
- To the treating healthcare team (need-to-know basis).
- By court order or legal summons.
- To public health authorities for notifiable/communicable diseases.
- **HIPAA (Health Insurance Portability and Accountability Act):** Federal law mandating national standards for protecting PHI (Protected Health Information). Covers confidentiality of both paper and electronic medical records.

Medical Audit

Definition

Medical Audit is a systematic review of medical care quality against established standards, aimed at improving patient outcomes and healthcare quality.

Objectives of Medical Audit

- Improve patient care quality and reduce errors.
- Support accreditation (NABH, JCI).
- Ensure accountability of medical and nursing staff.
- Identify deficiencies in care delivery.
- Formulate health policies and programmes.

Types of Medical Audit

Type	Description
Prospective	Reviews care standards before or during patient care to guide treatment.
Concurrent	Ongoing review during the patient's current episode of care.
Retrospective	Review of care quality after discharge, based on existing medical records.
Internal	Conducted by hospital staff. MRD staff perform quantitative analysis; medical personnel perform qualitative analysis.
External	Conducted by persons outside the treating team — audit committee or external specialists appointed by Director of Medical & Health Services.
Morbidity Audit	Reviews and records the main condition primarily responsible for the patient's need for treatment or investigation.

The Audit Cycle

Step 1: Select Topic → Step 2: Set Standards → Step 3: Collect Data → Step 4: Compare with Standards → Step 5: Identify Gaps → Step 6: Implement Changes → Step 7: Re-audit

Two Phases of Medical Auditing

- **Phase 1 — Quantitative:** Providing adequate records of performance as a basis of analysis (ensuring records are complete, present, and available).
- **Phase 2 — Qualitative:** Actual analysis of recorded data — reviewing clinical records and filed reports pertaining to professional work of the hospital.

Internal vs. External Audit Process

Internal Audit	External Audit
MRD staff collect all discharged patient records daily and arrange in chronological order.	Hospital administrator + selected clinicians hold monthly statistical meetings to discuss previous month's statistics.
Deficiencies of each record listed in a deficiency slip.	Medical Audit Committee (5–10 physicians/surgeons): frank, fearless, skilled, without prejudice.

Internal Audit	External Audit
Doctors complete deficient records during weekly doctors' conference / MRD room.	Director of Medical & Health Services selects specialists for periodic visits; they submit confidential reports to medical superintendent.

Common Deficiencies Found in Audit

1. Diagnosis without complete description; improper terminology.
2. Detailed diagnosis written inside the record but not on the face sheet.
3. Different final diagnosis recorded on face sheet vs. discharge summary.
4. Discharge notes incomplete or missing; only "D" written instead of full note.
5. No provisional diagnosis on the history and physical examination record.
6. Procedures/operations not recorded on the face sheet.
7. When provisional and final diagnosis are same, written as "same as above" (incorrect).
8. Result column not completed.
9. Signature of head of service/unit not found on the record.
10. Description of accident not written in RTA cases (e.g., head-on collision of cars, etc.).
11. Usage of improper medical forms.
12. Date and time of examination not written.

Solutions to Minimise Audit Deficiencies

Responsible Party	Responsibility
Medical Record Officer (MRO)	Establish written policies, standard forms, and guidelines on record completion. Maintain incomplete record control system. Create a Doctors' Conference Room within MRD. Submit regular reports on delinquent records with doctors' names.
Doctor	Document immediately after attending each patient. Senior doctors must supervise and verify junior doctors' entries for correctness and adequacy. Time spent on documentation should be considered part of clinical duties.
Nurse / Paramedical Staff	Complete all records of their respective services promptly and accurately.
Hospital Administrator	Provide transcription services and physical facilities in MRD. Institute rewards for prompt completion of records and penalties for delayed completion.

End of Summary — Medical Record Science I (BBAHM 202)